

Milestone 5 Write Up

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Risky Business: Quantitative Portfolio Management for Optimized Risk

Portfolio Resilience: Backtesting Asset Allocation Strategies During Financial Crises

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DATA 510 Data Science Capstone
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Title, Team, and Affiliations

Project Title: Risky Business: Quantitative Portfolio Management for Optimized Risk

Project Subtitle: Portfolio Resilience: Backtesting Asset Allocation Strategies During Financial Crises

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Repository: <https://github.com/awjackson00/data510-Quant-Back-Testing->

Abstract

This project looks at how different portfolio constructions perform during normal market conditions, growth periods, and financial crises. A lot of investors look mainly at total return, but that does not always show how much risk a portfolio is taking on. During crisis periods, portfolios that look strong in growth years can become fragile because of volatility, drawdowns, and changes in market behavior.

To study this, we built a reproducible backtesting workflow using historical daily market data from Yahoo Finance through the `yfinance` Python package. The data were frozen through June 28, 2026. We compared five constructed portfolios: Barbell, Mega-Cap Technology, Energy, Real Estate, and Fast-Moving Consumer Goods. We also compared them against benchmarks including the S&P 500 and market return minus the risk-free rate.

The project evaluates portfolios using cumulative return, annualized volatility, Sharpe ratio, Sortino ratio, crisis-period return, recovery behavior, and benchmark comparison. The main finding is that portfolio resilience cannot be judged by return alone. Mega-Cap Technology showed very strong upside during recovery and growth periods, but it also had larger swings during market stress. The Barbell strategy gave a useful defensive comparison, but it also has limits because the VIX is not directly investable. ARIMA forecasting was tested, but it did not provide conclusive results because daily return data are noisy and difficult to forecast. Overall, this project shows that stakeholders should evaluate portfolios using both return and downside-risk metrics.

Introduction: Problem and Research Questions

Many investors and analysts evaluate portfolios based on expected gains, recent performance, or exposure to popular sectors. This can be misleading because financial crises often reveal risks that are not obvious during normal market conditions. A portfolio can look strong during a growth period but perform poorly when volatility rises and markets start to fall.

This project focuses on risk-aware portfolio backtesting. Instead of only asking which portfolio made the most money, we wanted to evaluate which portfolios preserved value, limited downside risk, and performed better during market stress. We compare return potential, volatility, recovery behavior, Sharpe ratio, Sortino ratio, and crisis-period performance across different portfolio constructions.

The primary research question is:

How can portfolio optimization help stakeholders construct portfolios that preserve return potential while reducing drawdowns during periods of financial crisis and market uncertainty?

The supporting questions are:

1. Do drawdown-resistant portfolios experience lower gains during growth years compared to regular market indexes?
2. How does portfolio performance during crisis periods compare to performance during normal market conditions?
3. How are Sharpe and Sortino ratios affected during periods of uncertainty compared to periods of growth?
4. How do different asset classes and sectors, such as technology, energy, real estate, consumer staples, and defensive treasury-based assets, behave during periods of market stress?

These questions helped guide the project toward portfolio resilience rather than just return maximization. A portfolio is more resilient if it has reasonable return potential, lower volatility, stronger downside performance, and better risk-adjusted behavior during stressful markets.

Impact and Stakeholders

This project is designed for stakeholders who need to evaluate portfolio risk beyond simple return. The main stakeholders include institutional funds, risk-averse investors, portfolio managers, financial analysts, finance educators, investment clubs, risk managers, and students studying finance or portfolio analytics.

The practical value of this project is that it gives stakeholders a clearer way to think about portfolio construction. A portfolio with the highest return is not always the best portfolio for every investor. Some stakeholders may care more about stable, long-term gains and lower drawdowns than high-speed growth.

This project is especially useful for stakeholders who want to understand the tradeoff between upside potential and downside protection. A growth-heavy portfolio may perform very well during expansion periods, but it may also expose investors to larger losses during crises. A defensive portfolio may protect better during downturns, but it may also give up returns during strong markets.

The project is not meant to provide investment advice. The value is educational and analytical. The goal is to show how historical backtesting can help stakeholders think more critically about risk, diversification, asset allocation, and portfolio resilience.

Related Work and Background

This project builds on modern portfolio theory, risk-adjusted return analysis, backtesting, crisis-period analysis, and time-series modeling.

Modern portfolio theory argues that diversification can reduce risk by combining assets with different return patterns. However, crisis periods can challenge diversification because many assets may fall at the same time. When correlations rise during market stress, portfolios that seemed diversified during normal markets can become more fragile than expected.

Risk-adjusted performance metrics are important because return alone does not show how much risk a portfolio takes. The Sharpe ratio is a standard measure of return relative to total volatility. It compares the average return of a portfolio to the standard deviation of portfolio returns. A higher Sharpe ratio generally means a portfolio produced more return for each unit of total risk.

The Sortino ratio is similar, but it focuses more directly on downside risk. This is useful for this project because we care about drawdowns and negative returns during crisis periods. The Sortino ratio helps separate harmful volatility from upside movement.

We also tested ARIMA as a statistical time-series method. ARIMA was not used as a trading model. Instead, it was used as a simple baseline to see whether portfolio-level daily returns showed predictable patterns. The results were not conclusive, which makes sense because daily returns are noisy and often centered close to zero.

Important background sources for this project include Markowitz's work on portfolio selection, Sharpe's work on risk-adjusted returns, Sortino's downside-risk framework, Box and Jenkins' ARIMA time-series modeling framework, Taleb's *Antifragile*, Yahoo Finance, `yfinance`, and Tidy-Finance documentation.

Data and Data Engineering

The project uses historical daily market data from Yahoo Finance accessed through the `yfinance` Python package. The data scope was frozen through June 28, 2026. No additional market data were collected after this milestone, which helped keep the project consistent between the data summary, poster, final analysis, and write-up.

The project includes five constructed portfolios plus benchmarks.

Barbell Portfolio

The Barbell portfolio is based on a defensive strategy. In this project, the Barbell portfolio is 95% 10-year treasury exposure and 5% Volatility Index exposure. The goal was to test whether a defensive strategy could deliver adequate risk-adjusted returns during market stress.

Mega-Cap Technology Portfolio

The Mega-Cap Technology portfolio includes Apple, Microsoft, NVIDIA, Google, Amazon, Meta, Broadcom, and Tesla. This portfolio represents concentrated exposure to large technology and growth-oriented companies.

Energy Portfolio

The Energy portfolio includes ExxonMobil, Chevron, ConocoPhillips, EOG Resources, Schlumberger, NextEra Energy, Duke Energy, Constellation Energy, Southern Company, and Dominion Energy. This portfolio gives exposure to oil, utilities, and energy infrastructure.

Real Estate Portfolio

The Real Estate portfolio uses the Vanguard Real Estate ETF, VNQ. This gives the project a way to evaluate real estate exposure during different market environments, including periods with high interest rates.

Fast-Moving Consumer Goods Portfolio

The Fast-Moving Consumer Goods portfolio includes companies such as Walmart, Costco, Target, Procter & Gamble, Unilever, Kimberly-Clark, Coca-Cola, PepsiCo, Nestle, and General Mills. This portfolio represents consumer staples and defensive consumer demand.

Benchmarks

The benchmarks include the S&P 500 and market return minus the risk-free rate, labeled Mkt-RF. These benchmarks help show whether the constructed portfolios performed better, worse, or differently than broad market exposure.

The project pipeline follows this structure:

Ticker list and portfolio weights -> yfinance price pull -> adjusted prices -> daily returns -> weighted portfolio returns -> crisis labels -> performance metrics -> visualizations

The pipeline begins with portfolio definitions that connect tickers to portfolio weights. Historical adjusted prices are downloaded using `yfinance`. Those prices are converted into daily returns. Then the daily asset returns are weighted based on the portfolio construction rules. The final output is a set of portfolio-level return series that can be compared across crisis periods, recovery periods, and normal market periods.

The project uses pandas time-series dataframes instead of a full relational database. This works because the main unit of analysis is a daily return observation indexed by date. The main data path is:

**portfolio weights + asset returns -> weighted asset returns -> portfolio-level returns
-> crisis labels -> performance metrics**

The final analysis uses daily adjusted prices, daily asset returns, portfolio returns, benchmark returns, rolling Sharpe ratios, cumulative returns by regime, and risk-adjusted performance metrics.

Ethics and Responsible Use

All data used in this project are public financial market data. The project does not use personally identifiable information, private investor records, account-level data, or restricted financial information.

The main ethical risk is that someone could misread this project as investment advice. To avoid that, all results are framed as historical and educational analysis. The project does not recommend buying, selling, or holding any specific asset.

Another ethical risk is backtesting overconfidence. Backtesting can make a strategy look stronger than it would have looked in real time. It is easier to evaluate portfolios after we already know what happened in the market. To reduce this issue, we documented the portfolio construction rules and avoided claiming that these results will automatically hold in future crises.

Survivorship bias is also a limitation. The selected assets are securities that are known and available today. This may leave out companies or funds that failed, merged, or disappeared. As a result, the analysis may make historical portfolio construction look cleaner than it would have been in real time.

There is also a fair-comparison issue because different assets have different start dates. Some assets do not have full data across the entire period. The project handles this by rebalancing around available assets, but the limitation still matters when comparing early periods to later periods.

Methods and Evaluation

This project combines backtesting, risk analysis, benchmark comparison, crisis-period analysis, and ARIMA time-series modeling.

Each portfolio is evaluated over the frozen historical period using daily returns. Portfolio-level returns are calculated by applying portfolio weights to individual asset returns. This allows the portfolios to be compared across the same general market environments.

The project looks at several market regimes, including the Dot-com crash, the Global Financial Crisis, the post-crisis recovery and zero-interest-rate period, the COVID crash, the 2022 rate cycle, the March 2023 regional banking stress period, and the post-crisis period.

The project evaluates each portfolio using the following metrics:

- Cumulative return
- Annualized volatility
- Sharpe ratio
- Sortino ratio
- Crisis-period return
- Recovery behavior
- Benchmark comparison

These metrics were selected because they connect directly to the research question. Cumulative return shows upside potential. Annualized volatility shows instability. Sharpe ratio shows return compared to total volatility. Sortino ratio focuses more directly on downside risk. Crisis-period returns and recovery behavior show how portfolios respond under stress.

The project also uses rolling one-year Sharpe ratios. This helps show how risk-adjusted performance changes over time. A full-period Sharpe ratio can hide differences between regimes, so the rolling Sharpe ratio gives a better view of how performance changes during crashes, recoveries, and growth periods.

Cumulative return by regime is used to compare how portfolios behaved in specific market environments. This is important because one portfolio may perform well during a recovery but poorly during a crisis. The regime charts help show opportunity cost and downside protection more clearly.

ARIMA was also tested as a statistical time-series method on portfolio-level daily returns. The goal was not to predict the stock market perfectly. Instead, ARIMA was used as a basic stability check. The model did not produce conclusive results because daily portfolio returns are noisy and often centered close to zero. In several cases, ARIMA forecasts moved toward a flat near-zero line. This shows that simple time-series forecasting was not enough to explain daily return behavior in this project.

Results and Visualizations

The final results support the main idea that portfolio resilience should be measured in more than one way. Cumulative return, Sharpe ratio, Sortino ratio, volatility, and crisis-period performance each show a different side of portfolio behavior.

Figure 1. Rolling One-Year Sharpe Ratio by Portfolio

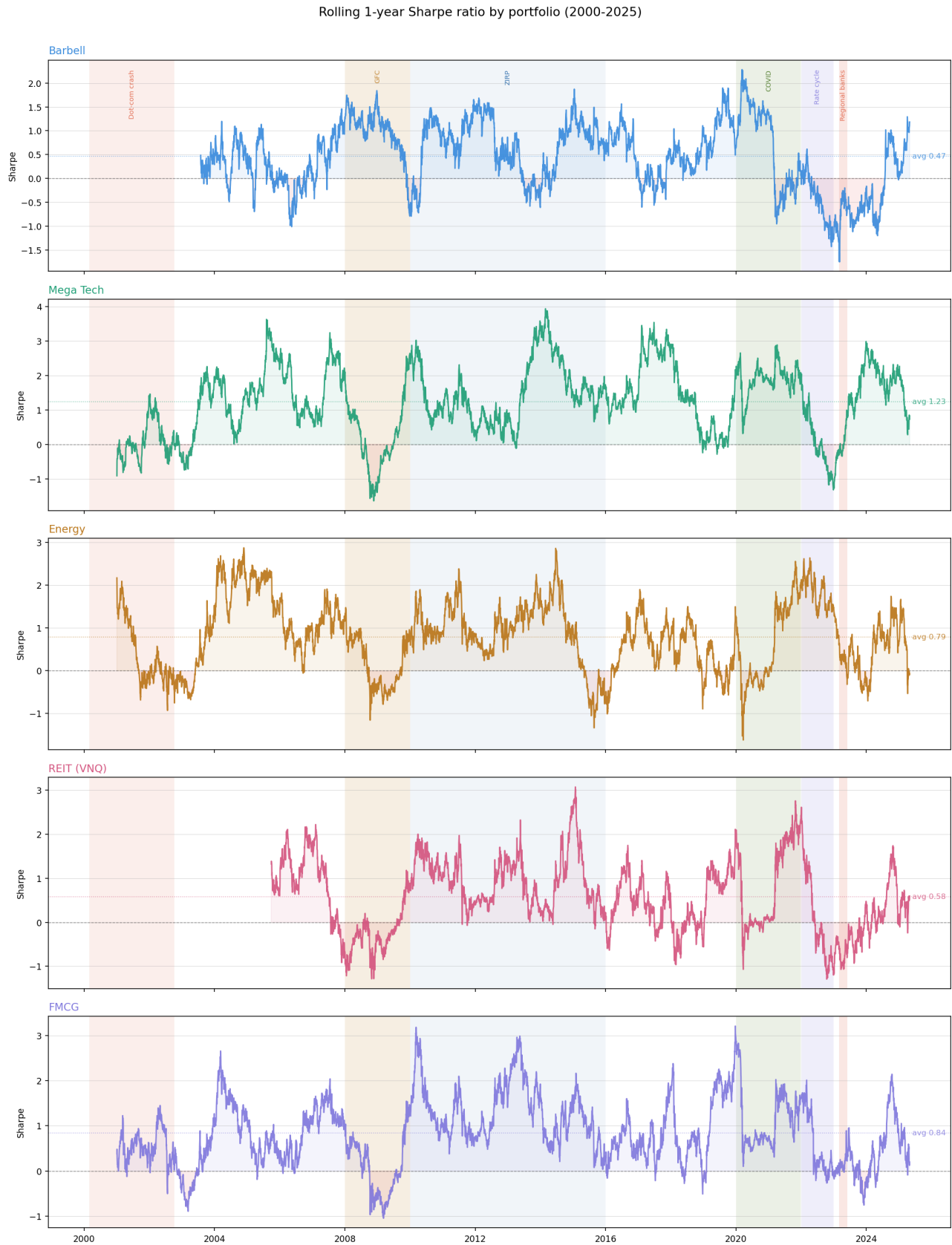


Figure 1 shows the rolling one-year Sharpe ratio for the Barbell, Mega-Cap Technology, Energy, Real Estate, and FMCG portfolios. The shaded regions mark different market regimes, including the Dot-com crash, Global Financial Crisis, recovery/ZIRP period, COVID period, rate cycle, and regional banking stress.

The figure shows that risk-adjusted performance changes a lot over time. Mega-Cap Technology had strong risk-adjusted performance in several growth and recovery periods. The Barbell portfolio had more modest average risk-adjusted performance, but it served as a defensive comparison. FMCG and Energy also showed positive rolling Sharpe behavior in several periods, but their performance depended heavily on the regime.

The main takeaway is that risk-adjusted performance is not stable across time. Market conditions matter. A portfolio that looks strong in one period may look weaker in another.

Figure 2. Benchmark Cumulative Returns by Regime

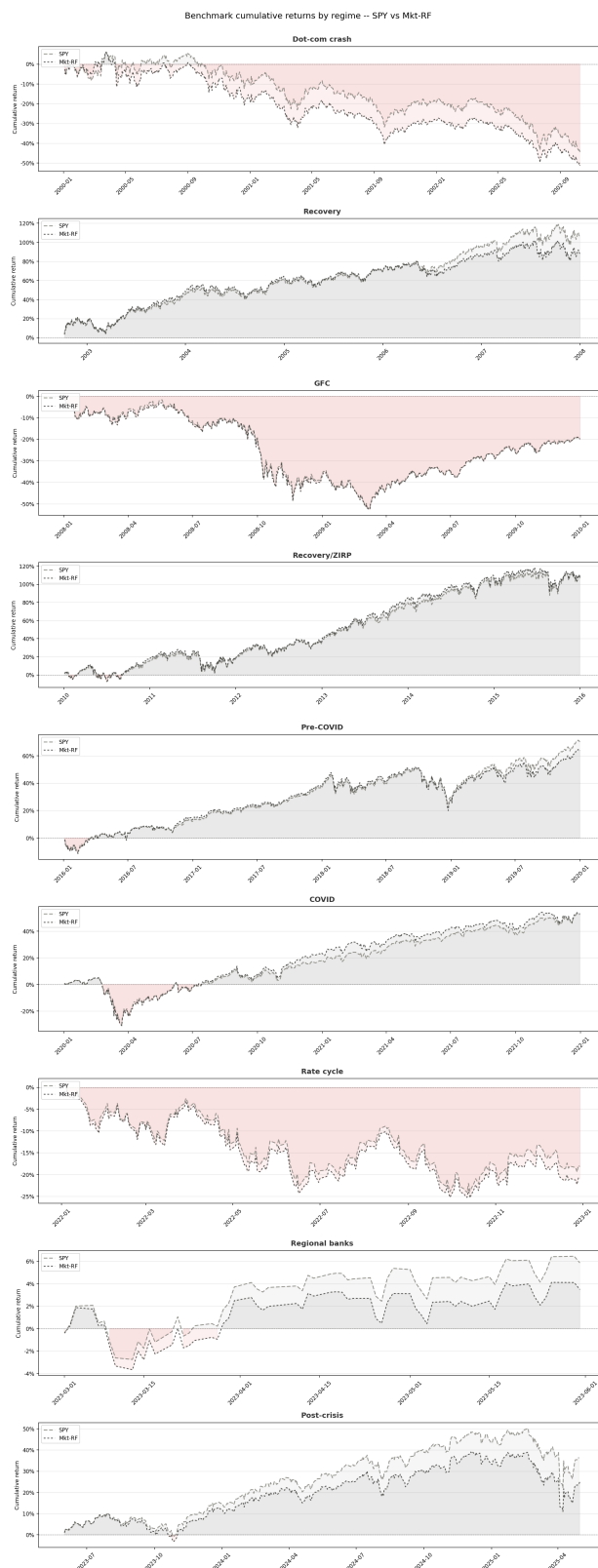


Figure 2 compares SPY and Mkt-RF across different market regimes. The benchmarks show large

losses during the Dot-com crash, the Global Financial Crisis, and the 2022 rate cycle. The COVID period shows a sharp decline followed by a strong recovery. The regional banking stress period was shorter and less severe for the broad benchmark series.

This figure gives context for the constructed portfolios. It shows that the broad market behaved very differently across regimes. Recovery periods produced strong cumulative gains, while crisis periods created large drawdowns and unstable performance.

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Figure 3. Cumulative Returns by Regime for Portfolios and Benchmarks



Figure 3 compares the five constructed portfolios against the benchmarks across different crisis and

recovery regimes. Mega-Cap Technology produced very large cumulative returns during recovery and expansion periods, especially after the Global Financial Crisis and before COVID. However, it also showed meaningful weakness during stress periods.

The Energy portfolio behaved differently depending on the crisis. It performed well in some periods but was not consistently defensive across all crises. This shows that sector resilience depends on the type of crisis.

The Barbell portfolio behaved more defensively in several stress periods, but it did not dominate in cumulative return. This supports one of the main tradeoffs in the project: drawdown-resistant strategies may reduce some downside exposure, but they can give up a lot of growth during long expansion periods.

The FMCG portfolio showed more stable behavior than high-growth technology in some stress environments. However, it did not match the upside of Mega-Cap Technology in strong growth periods.

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ARIMA Results

ARIMA was tested as a time-series forecasting method, but it did not produce conclusive results. Because the ARIMA output was not strong enough to support a final figure, it is discussed as a limitation rather than included as a visualization.

For daily portfolio returns, ARIMA often forecasted values close to zero. This makes sense because daily returns are noisy and usually centered around a small average return. The result does not mean ARIMA failed mechanically. It means that a simple ARIMA model did not reveal strong predictive structure in the portfolio return series.

For this reason, ARIMA should be treated as a basic statistical baseline, not as a trading model or as a main source of investment insight. The project's main findings are better supported by the benchmark comparison and cumulative return regime analysis.

Discussion, Limitations, and Future Work

The results show that portfolio resilience is not one single thing. Return, volatility, drawdowns, Sharpe ratio, Sortino ratio, and recovery behavior all matter.

Mega-Cap Technology produced strong upside during recovery and growth periods. This makes it attractive from a return perspective. However, it also had larger swings during stressful markets. This shows the risk of relying too much on growth-heavy assets.

The Barbell strategy gives a useful defensive comparison. Its purpose is not to maximize upside, but to preserve more stability during market stress. The results suggest that defensive construction can reduce some exposure to equity drawdowns, but it also comes with an opportunity cost. During long growth periods, defensive strategies can underperform growth-heavy portfolios.

Energy, Real Estate, and FMCG show that sector behavior depends on the crisis. Energy can perform well in some regimes and poorly in others. Real Estate is sensitive to interest-rate conditions. FMCG has a defensive consumer-staples logic, but it does not remove market risk completely.

The benchmark comparisons are also important. SPY and Mkt-RF show that broad market exposure can perform well during recoveries but can also suffer during major drawdown periods. Comparing constructed portfolios to benchmarks helps show whether a portfolio is actually resilient or just benefiting from a general market recovery.

There are several limitations to this project.

First, historical performance does not guarantee future performance. The project uses selected historical periods, and future crises may not behave the same way.

Second, backtesting can create false confidence. It is easier to build and evaluate portfolios after knowing what happened historically.

Third, survivorship bias may affect the results. The selected assets are securities that are available and known today. This may exclude companies or funds that failed or disappeared.

Fourth, some assets have shorter histories because of IPO dates or fund creation dates. Rebalancing around available assets helps the analysis continue, but it also means that portfolio composition changes over time.

Fifth, the VIX is not directly investable. This matters because the Barbell strategy uses VIX exposure as a volatility-sensitive component. In a real investment setting, an investable volatility ETF or options strategy would be needed.

Sixth, Yahoo Finance data may revise adjusted prices or change ticker availability. The project reduces this issue by freezing the data scope through June 28, 2026, but exact results could vary slightly if the data are pulled again later.

Seventh, ARIMA forecasting was inconclusive. It did not reveal strong predictive structure in daily portfolio returns.

Future work could improve the project by adding more formal max drawdown tables by regime, testing alternative crisis windows, replacing the VIX proxy with investable volatility products, adding transaction costs, testing different rebalancing schedules, and comparing optimized portfolios against the original constructed portfolios.

Conclusions and Recommendations

This project shows that portfolio resilience should not be judged by return alone. A portfolio with high returns can still expose stakeholders to large drawdowns and volatility. A defensive portfolio can reduce some downside risk, but it may also give up returns during long growth periods.

The strongest conclusion is that stakeholder priorities matter. Growth-oriented stakeholders may prefer portfolios with higher cumulative return. Risk-averse stakeholders may prefer portfolios with stronger downside behavior, lower volatility, or better risk-adjusted metrics.

The project also shows that the type of crisis matters. The Dot-com crash, Global Financial Crisis, COVID crash, rate cycle, and regional banking stress period affected portfolios differently. No single portfolio was automatically best across every regime.

The final recommendation is that stakeholders should use a multi-metric framework when evaluating portfolios. They should use cumulative return to understand upside potential, volatility to understand instability, Sharpe ratio to compare return against total risk, Sortino ratio to focus on downside risk, and crisis-period visuals to understand regime-specific behavior.

ARIMA and other forecasting models should be treated as supporting tools, not as the main decision engine. In this project, ARIMA did not provide conclusive results, which reinforces the need to rely on backtesting, risk metrics, and regime-specific comparisons.

Overall, this project provides a reproducible framework for evaluating portfolio resilience under historical market stress. The results are educational, not advisory, but they show why risk-aware evaluation is important for portfolio construction.

Reproducibility and References

The project repository is available at:

<https://github.com/awjackson00/data510-Quant-Back-Testing->

The final write-up and final poster should be committed under:

`deliverables/M5-final/`

The main dependencies are:

- *Python*
- `numpy`
- `pandas`
- `tidyfinance`
- `seaborn`
- `scikit-learn`
- `yfinance`
- `statsmodels`
- `matplotlib`

The main notebooks include:

- `notebooks/EDA Capstone.ipynb`
- `notebooks/SharpeRatioSheet.ipynb`
- `notebooks/portfolioConstruction.ipynb`

- `notebooks/ARIMA Time-Series Modeling.ipynb`

To reproduce the workflow, a reader should clone the repository, install the listed dependencies, run the EDA notebook to rebuild adjusted prices, daily returns, weighted portfolio returns, regime labels, and visualizations, then run the Sharpe/Sortino and ARIMA notebooks as needed.

The most important reproducibility point is the data freeze. The project uses historical data through June 28, 2026. Since Yahoo Finance data may revise adjusted prices or ticker availability, the frozen analysis window and documented code are necessary for keeping the poster, final write-up, and repository consistent.

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